

Data Visualization and Interpretation LAB

DVI Project

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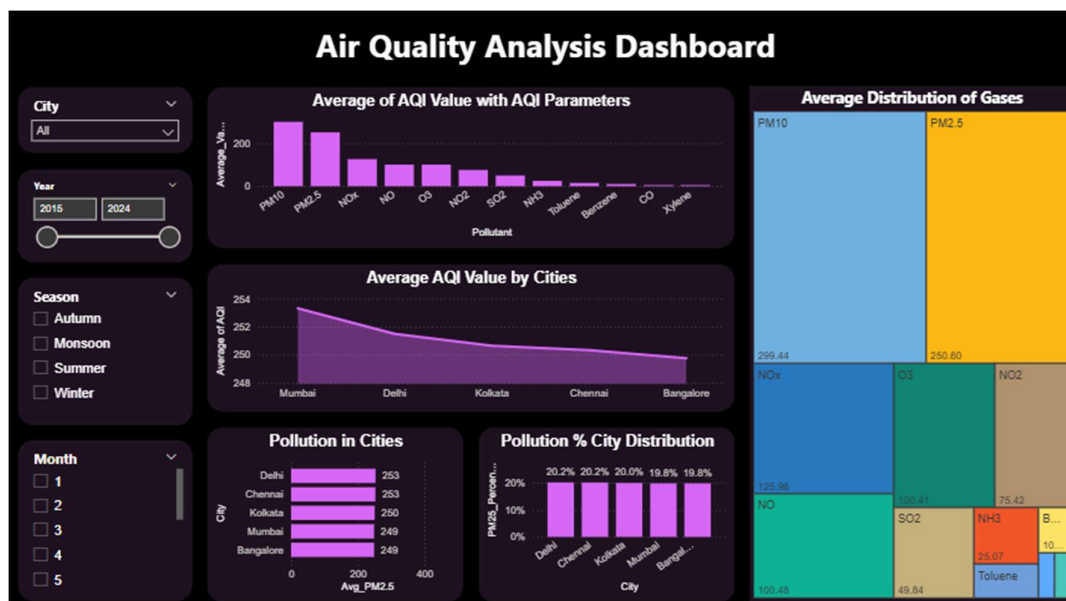
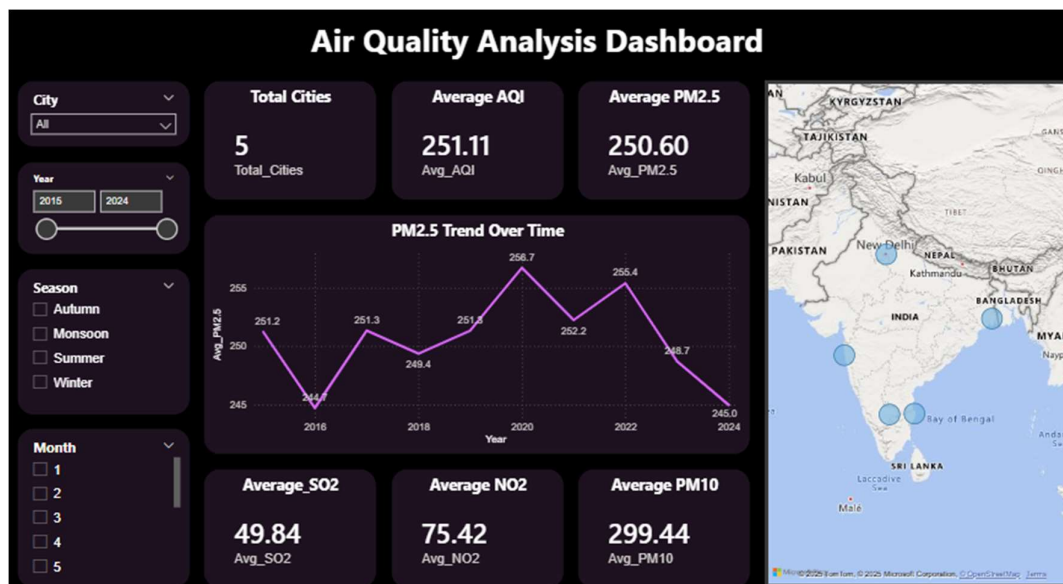
Air Quality Analysis insights

Report

This Air Quality Analysis Dashboard provides a holistic view of urban pollution levels, seasonal trends, and pollutant behavior across cities.

It helps identify high-risk locations, dominant pollutants, and temporal patterns impacting air quality.

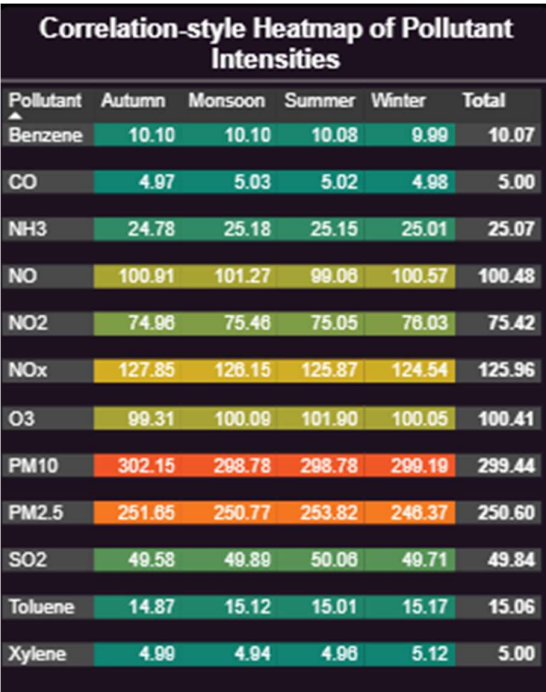
By visualizing clusters, trends, and comparisons, the dashboard enables quick environmental insights and supports data-driven decision-making for pollution control and policy planning.



PAGE 1: Exploratory Data Analysis (EDA) Insights

This page presents initial exploration of pollutant variation, distribution patterns, and relationships between key air quality indicators.

1. Population intensity varies across seasons

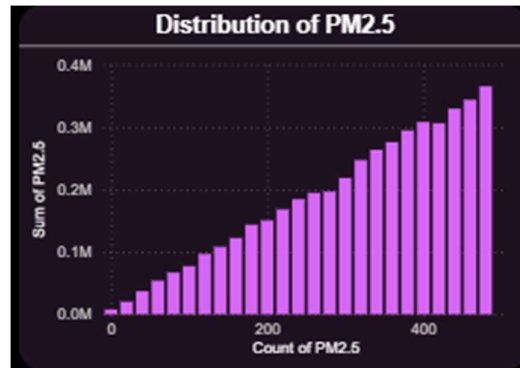


The heatmap reveals noticeable seasonal variations.
For example:

- PM10 and PM2.5 show peak average values in monsoon and summer
- NH3 and NOx display relatively stable seasonal behaviour

This highlights that seasonal weather dynamics significantly influence pollutant concentration.

2. PM2.5 distribution

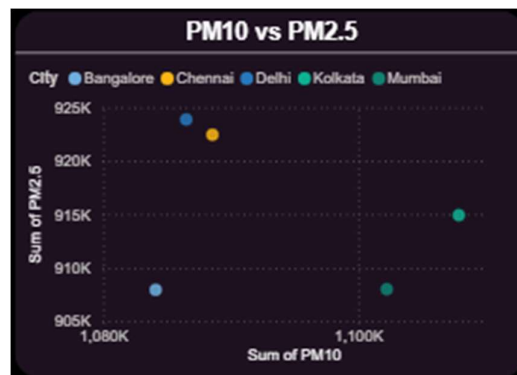


The histogram shows that the distribution of PM2.5 is:

- right-skewed i.e., most days are moderately polluted
- few instances of extreme pollution spikes

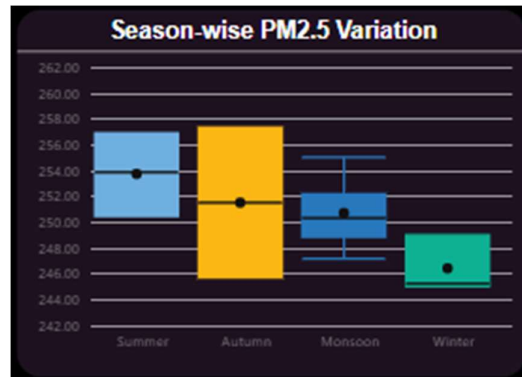
This suggests episodic pollution events instead of uniform pollution levels.

3. Strong pollutant relationship exists



- The scatter plot PM10 vs PM2.5 shows cities with high PM10 also show high PM2.5
- This indicates common emission sources such as traffic, combustion and dust.
- Delhi exhibits relatively higher PM2.5 values among selected cities indicating higher risk exposure.

4. Seasonal PM2.5 fluctuations demonstrate atmospheric impact



The boxplot shows:

- Autumn and Summer exhibit higher median PM2.5
- Winter shows lower variation and lower average

Seasonal meteorological behaviours like humidity, thermal trapping, and wind patterns likely drive these differences.

5. PM2.5 trend over years indicates inconsistent improvements



Line graph shows fluctuating improvement and worsening patterns:

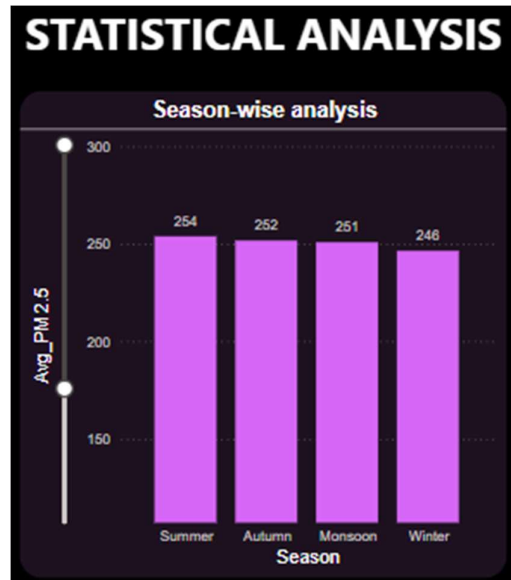
- Pollution rises sharply around 2020
- Declines gradually afterwards

This indicates that despite periodic improvement policies, pollution re-surges due to uncontrolled urban emissions.

PAGE 2: Statistical Analysis & K-Means Clustering Insights

This page combines inferential analysis with unsupervised learning to segment cities.

1. Seasonal PM2.5 difference is statistically meaningful

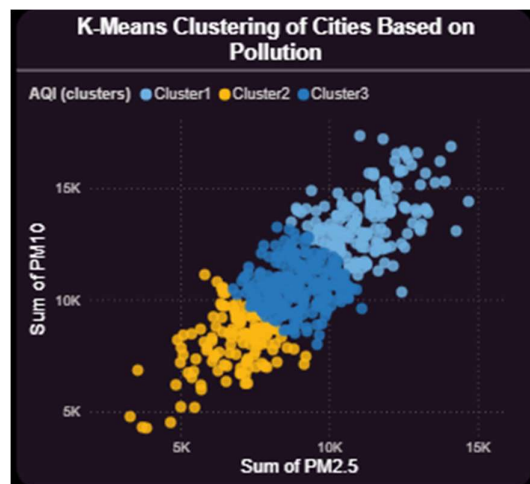


Bar comparison indicates:

- Summer ($\sim 254 \mu\text{g}/\text{m}^3$) and Monsoon ($\sim 261 \mu\text{g}/\text{m}^3$) have highest pollution
- Winter ($\sim 248 \mu\text{g}/\text{m}^3$) has the lowest

This pattern indicates "Season strongly influences air quality"

2. Cities form three natural pollution clusters (K-means)



- **Cluster 1: Low pollution cities**
 - Relatively cleaner, residential or regulated regions
- **Cluster 2: Moderate pollution cities**
 - Transitional cities with mixed industrial & urban activities
- **Cluster 3: High pollution cities**
 - Highly industrial/traffic dense zones with elevated health risks

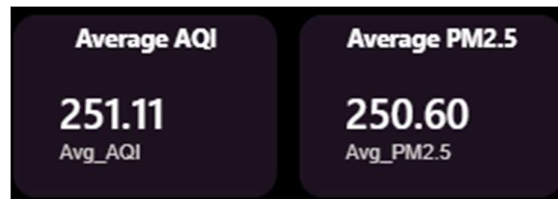
Positive slope between PM2.5 and PM10 shows that:

“Whenever fine particulate pollution rises, dust pollution rises as well, indicating shared emission origin.”

PAGE 3: Filter-interactive Dashboard Insights

This page highlights the effect of dynamic filtering by City, Year, Season, and Month.

1. Overall AQI and PM2.5 levels remain high (≈250–260 range)



Average pollutant levels exceed desirable safety ranges indicating:

- persistent air quality challenges
- universally polluted urban environment

2. Time trend shows fluctuations without structural improvement

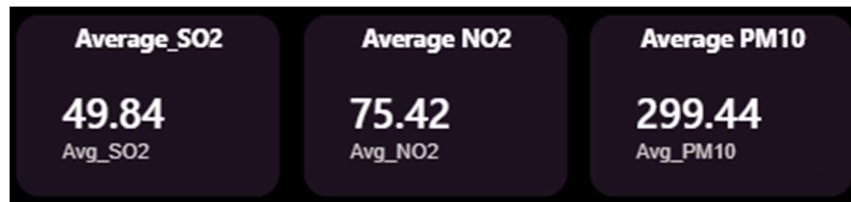


From 2015 to 2024:

- PM2.5 peaked around 2020–2022
- declines afterwards but remains high

This indicates **short-term interventions but no long-term mitigative stability**.

3. Pollutant wise analysis shows different dominance levels



Average pollutant values show:

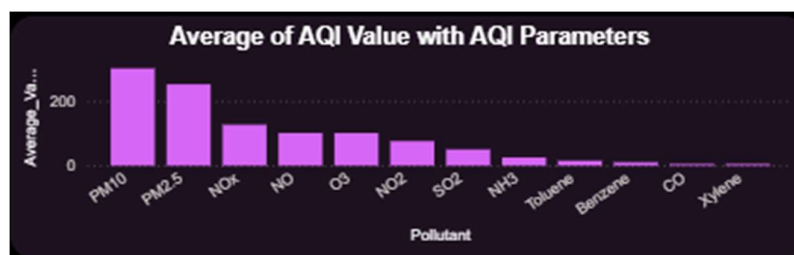
- SO2 and NO2 average ~ 50–75 $\mu\text{g}/\text{m}^3$
- PM10 remains significantly higher (~300 $\mu\text{g}/\text{m}^3$)

PM remains the most critical pollutant threat.

PAGE 4 — Comparison Dashboard Insights

This page compares parameters across cities, pollutant categories, and relative proportions.

1. PM10 and PM2.5 dominate pollutant load

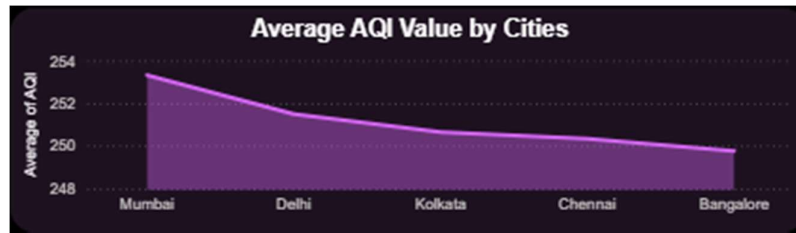


Treemap and bar analysis show:

- PM10 contributes the largest share
- PM2.5 second highest

This shows **particulate pollution as the main driver of AQI deterioration**.

2. City-wise AQI comparison reveals concentration of pollution stress



Mumbai, Delhi and Kolkata are among the highest AQI zones.

These cities show:

- higher vehicular emissions
- industrial density
- lower dispersal capacity

3. Pollution % contribution shows distribution is not uniform



Each city contributes around 18–20% of the total PM2.5 indicating:

- multiple cities facing similar magnitude of risk
- pollution is multi-city, not isolated to one location

Conclusion

- The dataset reveals **significant air quality challenges**, dominated by PM2.5 and PM10 pollutants.
- Seasonality impacts pollution, summer & monsoon typically peak.
- Year-wise trends indicate **cyclical improvement followed by worsening spikes**.
- Clustering validates that cities form **natural pollution severity categories**.

The project highlights the importance of integrated pollution monitoring, real-time analysis dashboards, and policy coordination for air quality management.